

Ahmed Manavi Alam

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [Google Scholar](#)

OBJECTIVE

Ph.D. candidate in Electrical Engineering at NC State University with 6+ years of research, teaching, and industry experience in machine learning, signal processing, and high-performance computing. Skilled in developing and deploying deep learning models using PyTorch and TensorFlow, with applications to wireless communication, radar, remote sensing, RF interference mitigation, room acoustics, and physics-aware ML. Industry experience includes two engagements at Bose Corporation, as a Machine Learning Research Engineering Intern and an Audio ML & DSP Research Co-op, and an internship at the High Performance Computing Collaboratory (HPC²) applying large-scale machine learning techniques.

EDUCATION

North Carolina State University Ph.D. in Electrical Engineering; Supervisor: Dr. Ali C. Gurbuz CGPA: 4.00/4.00	Raleigh, NC Oct 2026
Mississippi State University M.S. in Electrical and Computer Engineering; Supervisor: Dr. Ali C. Gurbuz CGPA: 4.00/4.00	Starkville, MS Aug 2024
Bangladesh University of Engineering and Technology B.S. in Electrical and Electronic Engineering; Supervisor: Dr. Nahid-Al-Masood	Dhaka, Bangladesh Apr 2019

WORK EXPERIENCE

Bose Corporation Machine Learning and DSP Research Engineering Intern	Framingham, MA May 2026–Aug 2026
– Developed an end-to-end deep learning system for blind room geometry estimation (length, width, height, and volume) from 6-microphone ambient audio, training CRNN models on STFT, phase, and inter-channel spatial (IPD, coherence) features to reach 15.32% dimension MAPE on held-out simulated rooms. – Engineered a large-scale data pipeline that synthesized 50K multi-channel RIRs across 300+ rooms and ran 75+ controlled training sweeps with automated leaderboards. – Designed a physics-informed multi-scale loss (dimensions, face areas, volume) that reduced dimension MAPE by 6.34% over a baseline CNN.	
North Carolina State University Graduate Research Assistant	Raleigh, NC Jan 2025–May 2026
– Developed RFI-Net, a U-Net-based deep learning framework for time-frequency semantic segmentation of radio frequency interference in passive microwave sensing, achieving AUC 0.9967 and reducing brightness-temperature error up to $5\times$ versus spectral-kurtosis baselines. [Paper] – Built a dual-polarized SDR-based L-band radiometer with onboard interference processing, reducing measurement uncertainty by 23%. [Paper]	
Bose Corporation Audio Machine Learning & DSP Research Co-op	Framingham, MA Jul 2024–Dec 2024
– Implemented a 1D ResNet model for room geometry estimation directly from room impulse responses, targeting out-loud (speaker) product applications, achieving an 8.48% performance improvement over baseline 1D CNNs. – Trained the model on synthesized RIR data and fine-tuned on real-world measurements, improving generalization to physical rooms and reducing the simulation-to-real gap. – Built and refined room acoustic simulation pipelines for sound propagation, absorption, and reverberation modeling, streamlining data generation and preprocessing across diverse room environments.	
Mississippi State University Graduate Research Assistant, Information Processing and Sensing Laboratory (IMPRESS)	Starkville, MS Sep 2023–June 2024
– Introduced HRSpecNET, a deep learning architecture for reconstructing high-resolution radar Micro-Doppler signatures (μ-DS) directly from 1-D complex time-domain signals, achieving a 3.48% improvement in human activity recognition accuracy over traditional STFT-based methods. [Paper]	

- Built a spectrum coexistence testbed in an anechoic chamber for wireless communication and passive sensing, with implementations in C/C++ and MATLAB, collecting more than 5000 real-world samples. [\[Paper\]](#) [\[GitHub\]](#)

High Performance Computing Collaboratory

Starkville, MS

Machine Learning Intern, Geosystems Research Institute (GRI)

June 2023–August 2023

- Implemented a deep learning-based computer vision model for precision agriculture that efficiently classified major crops using CNNs, Sentinel-2 data, and USDA's SCINet high-performance computing clusters.
- Led the implementation of advanced data preprocessing and vegetation phenology metrics, significantly improving data quality and classification performance. [\[ArcGIS StoryMap\]](#)
- Achieved 85% accuracy in high-resolution crop mapping by utilizing deep learning techniques, convolutional neural networks, and Sentinel-2 time series data.

Mississippi State University

Starkville, MS

Graduate Research Assistant, Information Processing and Sensing Laboratory (IMPRESS)

Aug 2021–May 2023

- Optimized a deep learning-based calibration framework for NASA's SMAP radiometer, achieving <0.5 K RMSE in antenna temperature estimation and reducing reliance on reference data by over 50%. [\[Paper\]](#)
- Deployed machine learning algorithms (Random Forest, XGBoost) to estimate Snow Water Equivalent (SWE) from SAR data collected by NASA's SWESARR instrument during the 2020 SnowEx campaign. [\[Notebook\]](#)

Daffodil International University

Dhaka, Bangladesh

Lecturer, Department of Electrical and Telecommunication Engineering

Sep 2019–Jul 2021

- Taught 24 undergraduate course sections to more than 460 students across core Electrical and Electronic Engineering subjects, blending theoretical instruction with hands-on laboratory training.

LEADERSHIP AND SYNERGISTIC ACTIVITIES

- **Reviewer**, IEEE Transactions on Aerospace and Electronic Systems [\[Publons Review Verification\]](#)
- **Reviewer**, IEEE Transactions on Radar Systems
- **Reviewer**, IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing
- **Reviewer**, IEEE Transactions on Geoscience and Remote Sensing
- **Contributor**, NASA Project (Title: Enhancing SMAP Radiometer Performance: Calibration and RFI Detection via Deep Learning, Award No: NNH23ZDA001N-SMAP) [Amount: \$493,000] [\[Award\]](#)
- **Contributor**, NSF Project (Title: AI-Enabled Spectrum Coexistence between Active Communications and Passive Radio Services: Fundamentals, Testbed and Data, Award No: 2030291) [Amount: \$516,000] [\[Award\]](#)
- **Contributor**, NSF Project (Title: INTeGrated Testbed Ensuring Resilient Active/Passive CoexisTence (INTERACT): End-to-End Learning-Based Interference Mitigation for Radiometers, Award No: 2332661) [Amount: \$750,000] [\[Award\]](#)
- **Session Manager**, IEEE IGARSS 2023 Conference (Technical Program: RFI Detection and Mitigation) [\[Technical Program\]](#)
- **Lead Organizer**, EEE Day 2019 (Supervising a team of 25 individuals to flawlessly manage every aspect, encompassing registration, coordinating with sponsors like Gigabyte Aorus, and facilitating prize distribution)

AWARDS AND HONORS

- **Winner**, AERPAW Autonomous Data Mule (AADM) Challenge
- **Fellowship**, 2025 Mentored Teaching Fellowship (Awarded by NC State University)
- **Finalist**, Aerospace and Electronic Systems Society Radar Challenge in IEEE RadarConf'24
- **Spring 2024 Graduate Research Symposium Winner** (Awarded by Mississippi State University) [\[News Article\]](#)
- **IEEE DySPAN 2024 NSF Student Travel Grant** (Awarded by NSF)
- **Finalist**, IEEE IGARSS 2022 Student Paper Competition (One of the top 10 papers among 250 submissions) [\[Finalists\]](#)
- **AERPAW Community Workshop Travel Award** (Awarded by NSF) [\[AERPAW\]](#)
- **Student Travel Award** (Awarded by Bagley College of Engineering, MSU)
- **Travel Assistance Grants for Graduate Students (TAGGS)** (Awarded by Office of the Graduate School, MSU)
- **National Academy of Sciences (NAS) Fellowship**
- **Government Merit Scholarship** (Awarded by Ministry of Education, Bangladesh)

SKILLS

- **Programming Languages and Software:** Python, C/C++, MATLAB, R, QGIS, ArcGIS, Simulink
- **Machine Learning Libraries:** Numpy, OpenCV, SciPy, Scikit-learn, Keras, PyTorch, TensorFlow
- **Additional Software:** LaTeX, Microsoft Excel, PowerPoint, Git
- **Cloud Computing and Database Management System:** AWS, USDA SCINet, MySQL

PUBLICATIONS

Published 6 peer-reviewed IEEE journal articles and 18 conference papers. Available at [\[Google Scholar\]](#).